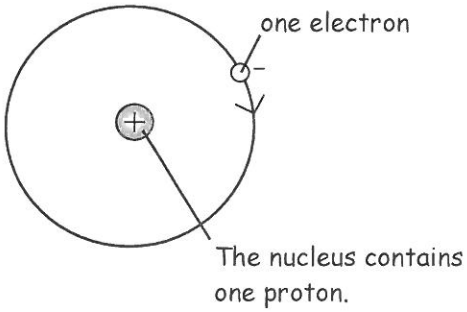
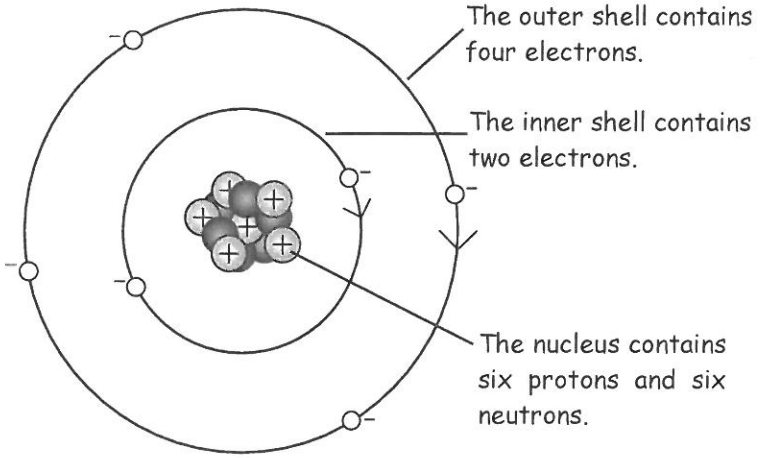
W.S.40. **Elements.** Name .....

An element is a pure substance that cannot be broken down into anything simpler. Everything on Earth is made from about one hundred different elements. An ATOM is the smallest particle of an element. They are much too small to be seen even with the most powerful microscope. Each element contains only one type of atom. Atoms have a NUCLEUS in the centre with ELECTRONS moving around it.

hydrogen atom



carbon atom



= positively charged proton

= neutron (no charge)

O^- = negatively charged electron

Atoms always have the same number of electrons and protons so that their overall charge is neutral (no charge).

Different elements have different numbers of protons in their atoms. The ATOMIC NUMBER is the number of protons that an atom contains. The smallest atom is hydrogen with an atomic number of one. Lead is one of the largest atoms with an atomic number of eighty two.

Exercise - Complete the missing words in the sentences below.

1) An _ _ _ _ _ _ _ cannot be broken down into anything simpler.

2) The smallest particle of an element is called an _ _ _ _

3) The _ _ _ _ _ _ _ is in the centre of an atom.

4) Electrons have a _ _ _ _ _ _ _ _ charge.

5) Protons have a _ _ _ _ _ _ _ _ charge.

6) The atomic number is the number of _ _ _ _ _ _ _ in an atom.

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